

# INMO(25<sup>th</sup>) – 2010

Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
  - Ruler and compass allowed.
  - Answer all the questions.
1. Let ABC be a triangle with circum circle  $\Gamma$ . Let M be a point in the interior of triangle ABC which is also on the bisector of  $\angle A$ . Let AM, BM, CM meet  $\Gamma$  in  $A_1, B_1, C_1$  respectively. Suppose P is the point of intersection of  $A_1C_1$  with AB; and Q is the point of intersection of  $A_1B_1$  with AC. Prove that PQ is parallel to BC.
  2. Find all natural numbers  $n > 1$  such that  $n^2$  does not divide  $(n-2)!$ .
  3. Find all non-zero real numbers x, y and z which satisfy the system of equation:  

$$(x^2 + xy + y^2)(y^2 + yz + z^2)(z^2 + zx + x^2) = xyz$$

$$(x^4 + x^2y^2 + y^4)(y^4 + y^2z^2 + z^4)(z^4 + z^2x^2 + x^4) = x^3y^3z^3.$$
  4. How many 6-tuples  $(a_1, a_2, a_3, a_4, a_5, a_6)$  are there such that each of  $a_1, a_2, a_3, a_4, a_5, a_6$  is from the set  $\{1, 2, 3, 4\}$  and the six expressions  $a_j^2 - a_j a_{j+1} + a_{j+1}^2$  for  $j = 1, 2, 3, 4, 5, 6$  (where  $a_7$  is taken as  $a_1$ ) are equal to one another?
  5. Let ABC be an acute angled triangle with altitude AK. Let H be its orthocentre and O be its circumcentre. Suppose KOH is an acute angled triangle and P its circumcentre. Let Q be the reflection of P in the line HO. Show that Q lies on the line joining the mid point of AB and AC.
  6. Define a sequence  $\langle a_n \rangle_{n \geq 0}$  by  $a_0 = 0, a_1 = 1$  and  $a_n = 2a_{n-1} + a_{n-2}$  for  $n \geq 2$ .
    - (a) For every  $m > 0$ ,  $0 \leq j \leq m$ , prove that  $2a_m$  divides  $a_{m+j} + (-1)^j a_{m-j}$ .
    - (b) Suppose  $2^k$  divides n for some natural numbers n and k. Prove that  $2^k$  divides  $a_n$ .

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